# Lords Institute of Engineering & Technology

Department of Computer Science and Engineering

## MAJOR-PROJECT PRESENTATION - I

# SecureAuthX: A Cross-Browser Extension for Secure Password Management and Website Security Analysis


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THE DIRECTION

# One extension, two layers of browser security



## Website security

**Current direction:** website security analysis with phishing detection.

**Current direction:** privacy and fingerprinting analysis inside the browser extension.



## Credential protection

**Proposed direction:** local-first password management designed around an encrypted local vault protected by a master password.

Password generation and strength evaluation support secure password management.



## Local by design

Core password data is intended to remain local, with no cloud storage for password management.

THE PROBLEM

# Users face credential risk and website risk at the same time

Weak and reused passwords increase account compromise risk.

Phishing websites imitate legitimate destinations.

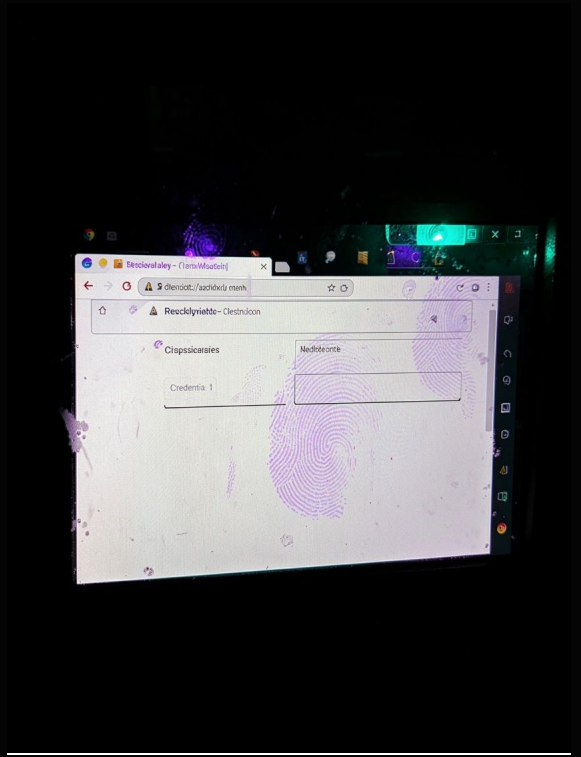
Fingerprinting can identify or track users across browsing contexts.

Users may submit credentials to unsafe form destinations.

Password management and website security are often handled by separate tools.

PROBLEM → RISK

Fragmented protection leaves the credential and browsing context disconnected.



THE GAP

# Existing tools solve pieces of the problem



**Conventional password managers** protect credentials but may not evaluate website context.



**Password generators** improve password creation but do not analyze destinations.



**Browser-based password management** can lack integrated privacy analysis.



**Separate security and privacy extensions** fragment the user experience.



**Some password-management approaches** depend on external or cloud infrastructure.

# SecureAuthX connects credential protection to website context



**Proposed encrypted local vault:** master-password-based protection

**Proposed password security:** generation, strength, and reuse analysis

**Proposed secure autofill:** user-initiated credential filling

**Current analysis direction:** phishing and unsafe-destination analysis

**Current analysis direction:** fingerprinting and tracking-signal analysis

**Design principle:** local-first processing within the extension

Conceptual flow: Password Management + Website Security + Privacy Analysis → SecureAu

PHASE I

# Six objectives define the Phase-I direction

01

Develop a secure browser-based password management system.

02

Protect stored credentials using an encrypted local vault.

03

Provide secure password generation and strength evaluation.

04

Detect phishing and unsafe website or credential-submission conditions.

05

Analyze privacy and browser fingerprinting signals.

06

Minimize dependence on external services while assisting users during browsing.

FUNCTIONAL SCOPE

# Four modules turn signals into practical protection



## Password Management

PROPOSED

Encrypted local vault · Master password · Password generation · Strength/reuse analysis · User-initiated autofill



## Website Security

HTTPS · Form destination · Phishing/typosquat · Identity-provider awareness



## Privacy Analysis

Fingerprinting signals · Tracking/privacy analysis · Browser indicators



## Security Scoring

Security score · Privacy score · Phishing safety · Overall trust assessment

SYSTEM DESIGN

# Proposed architecture connects page signals to secure user action

Web Page →

Analysis Engine → Website Security

Content Script / Page Hook →

Analysis Engine → Privacy/Fingerprinting Analysis

SecureAuthX Analysis Engine

Analysis Engine → Password Management

**Password Management → Popup Interface → Encrypted Local Vault**

**Password workflow:** Master Password → Unlock Vault → Select Website/Account → Evaluate Website Context → User-Initiated Autofill → Fill Required Credential

No cloud server is part of the proposed architecture.

IMPLEMENTATION + SECURITY

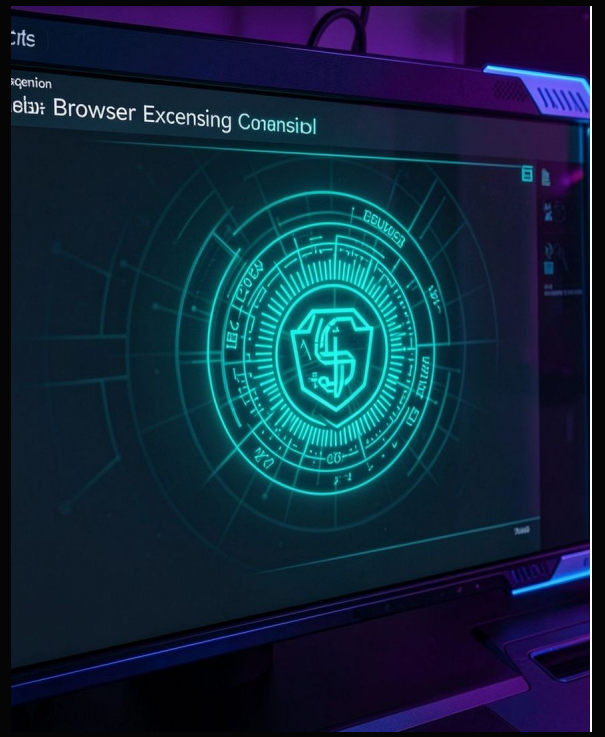
# The security stack keeps sensitive data local

## Technology stack

Current implementation/testing focus:
Chrome Extension / Manifest V3
HTML · CSS · JavaScript
Web Crypto API · AES-256-GCM · PBKDF2-SHA-256
Browser local storage
Unit Testing + Playwright E2E Testing

## Security principles

Master password is never stored
Proposed vault data is encrypted before persistent storage
Decrypted data remains in memory only while unlocked
Autofill is user initiated; only required credentials cross the messaging boundary
No cloud storage for core password data and no unnecessary new permissions



CONCLUSION

# SecureAuthX establishes a local-first foundation

SecureAuthX combines password management with website security analysis so users can protect credentials locally while identifying phishing, security, and privacy risks during browsing.

## Phase-I / current development


Website security and privacy analysis


Password security functionality


Privacy/fingerprinting analysis


Local-first security architecture


Foundation for encrypted vault and secure autofill

## Future scope

- Autofill compatibility and hardening
- Expanded cross-browser validation
- More extensive security testing
- Improved vault usability and UI/UX refinement
- Additional security/privacy detection capabilities

Secure credentials. Safer browsing. Privacy by design.